Research Methodology-I

Scheme

Course Title: Research Methodology-I										
Semester	VI	Teaching Scheme				Evaluation Scheme				
						Theory			Practical	
Term	Even	Th.	Tu.	Pr.	Credits	TAE	CAE	ESE	INT	EXT
Course Category	RM	-	-	4 Hrs.	2	10	15	-	10	-
Course Code										
Teaching Mode	Offline	2 Hrs.			Total	0			35	
Duration of ESE	-					35 Marks				

Course Objectives

1. To develop a foundational understanding of scientific research, its evolution, characteristics, and significance in modern inquiry.

2. To enable students to design, plan, and execute research by selecting appropriate methodologies, models, and data collection techniques.

3. To equip learners with skills in data analysis using statistical tools and to evaluate research findings with sound interpretation

Course Outcomes

CO1 Identify and define research problems by understanding scientific research processes and experimental design.

CO2 Formulate appropriate research designs and models using inductive and deductive reasoning.

CO3 Select suitable data collection methods and sampling techniques based on the research objective.

CO4 Apply statistical tools for data analysis and perform hypothesis testing using both parametric and non-parametric tests.

WHAT IS RESEARCH?

The word **research** is composed of two syllables, **re and search.**

- **re** is a prefix meaning **again, anew or over again**
- **search** is a verb meaning to **examine closely and carefully**, to **test and try, or to probe**

Definition: Research is undertaken within most professions. More than a **set of skills**, it is a **way of thinking**: examining critically the various aspects of professional work. It is a **habit of questioning** what you do, and a **systematic examination** of the **observed** information to find answers with a view to instituting appropriate changes for a more effective professional service.

Key Aspects of Research

- **Systematic Process**: It involves a series of structured steps, from defining a problem to drawing conclusions.
- **Goal-Oriented**: Aims to establish facts, principles, or new understandings.
- **Movement**: Often described as moving from the known to the unknown

Evolution of Scientific Inquiry

Best Research Contributions by Indians (2nd–20th Century)

2nd Century Charaka – Medicine (Ayurveda)

•Authored *Charaka Samhita*.

•Systematic study of human anatomy, diseases, diagnosis, and treatment.

•One of the earliest examples of **medical research based on observation and experimentation**.

6th Century Varahamihira – Astronomy and Earth Sciences

- Combined astronomy, meteorology, and geography.
- Studied rainfall patterns, earthquakes, and planetary motion.
- Early example of **interdisciplinary research**.

12th Century Bhaskara II (Bhaskaracharya) – Mathematics and Astronomy

- Early ideas similar to **calculus (rates of change)**.
- Advanced work in algebra, trigonometry, and planetary motion.
- Strong blend of **theory and mathematical modeling**.

19th Century Jagadish Chandra Bose – Physics and Biology

- Research on **radio waves** before Marconi.
- Proved that **plants respond to stimuli** like living organisms.
- Developed sensitive scientific instruments.
- Early **experimental and interdisciplinary research**.

5th Century Aryabhata – Mathematics and Astronomy

- Proposed that the Earth rotates on its axis.
- Calculated the value of π (pi) accurately. Explained solar and lunar eclipses scientifically.
- Major contribution to theoretical scientific inquiry.

7th Century Brahmagupta – Mathematics

- Defined rules for **zero and negative numbers**.
- Advanced algebraic equations.
- Foundation for modern mathematics and computation.

16th–17th Century Indian Metallurgists and Engineers

- Production of **wootz steel**, later used to make Damascus steel.
- Advanced metal extraction and alloy techniques.
- Example of **applied and materials research**.

Early–Mid 20th Century Srinivasa Ramanujan – Mathematics

- Groundbreaking work in **number theory, infinite series, and equations**.
- His theories are still used in modern mathematics and physics.
- Example of **pure theoretical research**.

Evolution of Scientific Inquiry

Scientific inquiry evolved from:

- **Speculation and philosophy**
- **Observation-based explanations**
- **Experimental verification**
- **Hypothesis testing**
- **Evidence-based and interdisciplinary research**

Modern scientific inquiry emphasizes:

- Critical thinking
- Data analysis
- Ethical research practices
- Replicability

Science and Research

- Overview of Science and Scientific Research
- Science is a structured system of knowledge that explains **natural phenomena through observation, experimentation, and logical reasoning**. It emphasizes objectivity, empirical evidence, and reproducibility to establish reliable principles and laws.
- Scientific research operationalizes science by applying **systematic methods** to **generate new knowledge, validate existing theories, and solve practical problems**.
- Scientific research integrates curiosity with methodological rigor.
- It involves identifying problems, formulating hypotheses or objectives, designing appropriate methodologies, collecting and analyzing data, and drawing valid conclusions. Together, science and scientific research drive innovation, technological advancement, and evidence-based decision-making across disciplines, including natural sciences, engineering, medicine, and social sciences.

Science

Science is a systematic and organized body of knowledge acquired through **observation, experimentation, and logical reasoning**. It seeks to explain **natural phenomena, establish cause–effect relationships, and develop general laws and principle**s that are verifiable and reproducible.

Key Features of Science:

- Systematic and organized
- Based on empirical evidence
- Objective and verifiable
- Predictive and explanatory
- Dynamic and evolving

Scientific Research

Scientific research is a systematic, objective, and logical process of inquiry aimed at discovering, interpreting, or revising knowledge.

- Systematic investigation
- Discovery and verification of knowledge

Characteristics of Research

- Systematic
- Objective
- Empirical
- Replicable
- Ethical

Characteristics of Research

To qualify as research, a process must have certain characteristics. These characteristics are:

Controlled: The study is set up to minimize the effects of external factors on the relationship between variables, particularly when exploring causality.

Valid and verifiable: The conclusions drawn from the findings are correct and can be confirmed by the researcher and others through replication.

Empirical: Conclusions are based on hard evidence gathered from real-life experiences or observations, not just theory or logic.

Critical: The procedures and methods used undergo careful and precise scrutiny to ensure the process is free from drawbacks and can withstand criticism.

Rigorous: The procedures followed to find answers are relevant, appropriate, and justified, with scrupulous attention to detail.

Systematic: The investigation follows a certain logical sequence, with different steps proceeding in an orderly manner.

Types of Research

A. Based on Purpose

- Basic (Fundamental) Research
- Applied Research
- Action Research

B. Based on Method

- Experimental Research
- Descriptive Research
- Historical Research
- Survey Research

C. Based on Data

- Qualitative Research
- Quantitative Research
- Mixed Method Research

Basic (Fundamental) Research: Aimed at generating new knowledge without immediate practical application.

Applied Research: Focused on solving specific practical problems.

Experimental Research: Involves controlled experiments to establish cause–effect relationships.

Descriptive Research: Describes characteristics of a phenomenon or population.

Analytical Research: Uses existing data to analyze and interpret relationships.

Qualitative and Quantitative Research: Based on non-numerical and numerical data, respectively.

Types of Research

Descriptive vs. Analytical Research

Descriptive Research

Meaning:

Describes the present situation or facts as they exist. The researcher does **not control variables**, only observes and reports.

Key points (Short):

- Describes "what is happening"
- Uses surveys, questionnaires, observations
- Also called **ex post facto research**

Example:

- A survey studying **shopping frequency of customers in a mall**
- Study on **students' preferred online learning platforms**

Analytical Research

Meaning:

Uses **already available data or facts** and analyzes them to make a **critical evaluation or interpretation**.

Key points (Short):

- Uses existing data
- Focuses on analysis and interpretation
- Explains "why" or "how"

Example:

- Analyzing **past sales data** to identify reasons for a decline in profit
- Studying **existing exam results** to evaluate student performance trends

Types of Research

Applied vs. Fundamental (Basic) Research

Applied Research

Meaning:

Conducted to solve an **immediate practical problem** faced by society or industry.

Key points (Short):

- Problem-oriented
- Practical use
- Immediate application

Example:

- Market research to **improve product sales**
- Research to **reduce defects in iron ore pellets**
- Study on **customer satisfaction for a company**

Fundamental (Basic) Research

Meaning:

Aims to **increase knowledge** and develop **theories or principles**, without immediate practical application.

Key points (Short):

- Knowledge-oriented
- Builds theory
- Long-term importance

Example:

- Research on **human behavior patterns**
- Study of **properties of materials**
- Research in **pure mathematics or physics**

Types of Research

Quantitative vs. Qualitative Research

Quantitative Research

Meaning:

Deals with **numerical data** and measurements. The results are expressed in **numbers, percentages, graphs, or statistics**.

Key points (Short):

- Focuses on measurement of quantity or amount
- Deals with data that can be expressed numerically
- Uses tools like statistics, surveys, experiments, and numerical analysis
- **Example:**
- Survey showing **80% students prefer online classes**
- Measuring **production rate, cost, or efficiency**
- Measuring customer satisfaction using a 1–5 rating scale

Qualitative Research

Meaning:

- Deals with **qualities, opinions, feelings, attitudes, and behavior**.
- It explains **why and how people think or behave**.

Key points (Short):

- Non-numerical data
- Focuses on motives and perceptions
- Uses interviews, discussions, projective techniques

Example:

- Studying **why students feel stressed during exams**
- Research on **customer satisfaction and opinions**

Need of Research

Research is needed to **generate knowledge, solve problems, and support decision-making**. Its importance can be summarized as follows:

- **To discover new knowledge:** Helps in developing new ideas, concepts, and theories.
- **To solve practical problems:** Provides solutions to real-life issues in industry, society, and technology.
- **To improve existing practices :** Enhances methods, processes, products, and systems through systematic study.
- **To support decision-making:** Supplies reliable data and evidence for informed decisions.
- **To verify and validate facts :** Confirms existing information through observation and experimentation.
- **To understand relationships between variables:** Explains cause–effect relationships in scientific and social phenomena.
- **To aid planning and policy formulation:** Helps governments and organizations design effective policies and strategies.
- **To promote innovation and development:** Drives technological advancement and economic growth.

Identification of Problem

Meaning: Selecting a clear, specific, and researchable problem that needs investigation.

Sources of Research Problems:

- Literature review
- Personal experience
- Social issues
- Gaps in existing studies
- Emerging trends

Assessing Problem Status

Assessing problem status means clearly understanding and evaluating a research problem before starting the study. It helps in deciding what to study, why to study, and how to study.

- ✓ **Problem identification:** Clearly defines the research problem.
- ✓ **Understanding the nature of the problem :** Determines whether the problem is theoretical or practical.
- ✓ **Feasibility analysis :** Checks availability of data, time, cost, and resources.
- ✓ **Scope definition :** Fixes the boundaries and limitations of the study.
- ✓ **Setting objectives ;** Helps in framing clear and achievable research objectives.
- ✓ **Selection of research method ;** Assists in choosing appropriate tools and techniques.
- ✓ **Avoids duplication :** Ensures the problem has not already been solved.
- ✓ **Improves research quality :** Leads to systematic and meaningful research outcomes

Formulating Objectives

Formulating objectives means **clearly stating what the research intends to achieve**. Objectives guide the entire research process and keep the study focused.

Key Points :

- **Clarify the purpose of research :** Defines what the study aims to accomplish.
- **Convert problem into goals :** Transforms the research problem into specific targets.
- **Provide direction to the study :** Helps in selecting methodology, tools, and analysis techniques.
- **Define scope and limitations :** Prevents unnecessary deviation from the topic.
- **Make research measurable :** Objectives should be clear, specific, and achievable.
- **Help in evaluation of results :** Findings are compared against stated objectives.

Objectives

Objectives specify what the researcher intends to achieve.

Types:

General objectives : Overall aim of the study

Examples:

- To develop an efficient machine learning model for prediction.
- To develop an AI model for disease prediction.

Specific Objectives: Detailed and measurable aims

Examples:

- To collect medical datasets
- To perform feature selection and normalization
- To train deep learning models for diagnosis
- To validate predictions using performance metrics

Characteristics of Good Objectives:

- Clear and precise
- Measurable
- Achievable
- Relevant to the problem

Research Design (Experimental)

Research Design:

A blueprint or plan for conducting research.

Experimental Design Includes:

- Selection of variables (independent, dependent, control)
- Selection of sample
- Tools and instruments
- Procedure for data collection
- Control of extraneous variables

Common Experimental Designs:

- Pre-test and post-test design
- Control group design
- Randomized design

Actual Investigation

This is the **execution stage** of research.

Steps Involved:

- Data collection
- Application of tools/instruments
- Observation and experimentation
- Recording data accurately
- Ensuring ethical standards